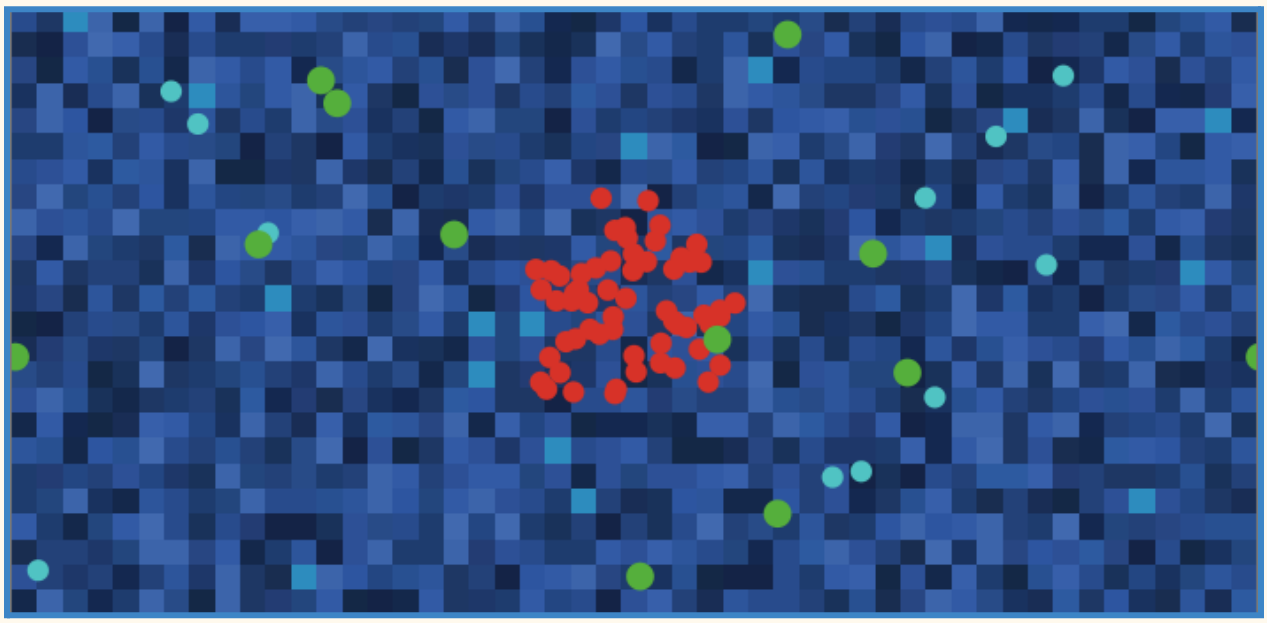
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TUMOR-IMMUNE MICROENVIRONMENT

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INTRODUCTION

This report explains the findings of a simple *tumor-immune microenvironment (TME)* simulation which comprises agents representing the actors interacting with the environment. These agents provide a representation of the behaviour of *tumoral cells, macrophages, cytotoxic T* cells who interact with the surrounding environment that provides *oxygen, nutrients and cytokines* to these actors. In this simplified *Multi Agent System (MAS)* model, each actor perceives the environment and takes Actions based upon the knowledge obtained from their surroundings.

For example, macrophages can inhibit or support tumor cells growth while cytotoxic T cells may kill nearby tumor cells. At the same time, the environment directly influences the evolution of the simulation: nutrient limitation or oxygen limitation may alter the proliferation of cells and increase their death rate. This report shows the design decisions, the system's architecture and the experiments that have been carried out along with their results.

HYPOTHESES AND RESEARCH QUESTIONS

1. OXYGEN DEPRIVATION:

- The environment is set with a low level of oxygen in order to answer the following research question: “How does oxygen deprivation affect cell growth and their death rate & the immune responsiveness of every cell?”

  — “If global oxygen supply is set to ‘0’, tumor cells will lose energy, experience more hypoxia, increase resistance, and the tumor population should decrease compared with normal ‘0.15’ oxygen supply —

2. HIGH IMMUNE STRENGTH:

- The environment is set with a high immune strength in order to answer the following question: “How does high immune strength affect cell growth and their death rate & the immune responsiveness of every cell?”

  — “The global immune strength will increase the tumor cell killing and reduce the final tumor population. The cell damaged is directly multiplied by immune strength —

SYSTEM COMPONENTS

The system is composed of the following main components:

- **Agent (in NetLogo Framework defined as Breeds):**
  - Tumor Cells
  - T-Cells
  - Macrophages
  - Environment-Agents
- **Resources (in NetLogo Framework defined as Patch-Owned variables):**
  - Oxygen
  - Nutrients
  - Cytokine
  - Vessel

Note: Agents and Resources are initialized during an initialization phase.

- **Environment (contains Agents and Resources):** In the Environment Definition phase, Blood Vessels are designated as sources of oxygen and nutrients, which will then be diffused, as well as cytokine, in the surrounding environment. Cytokines are modeled with a finite lifetime and consequently undergo temporal decay. Furthermore, the environment incorporates a mechanism for detecting hypoxia, defined as insufficient oxygen availability, and produces warning signals whenever such conditions arise.
- **Rules:** The main simulation loop is guided by rules.

AGENTS OVERVIEW

Tumor Cell

- **Internal state:** Energy, Resistance, Inbox (message from nearby cells).
- **Perception:** Oxygen, Nutrients, Macrophage support, Hypoxia signals, Immune Attacks
- **Action/Rules:** Consume resources, +/- Resistance, Release Cytokines, Suppress T cells, Divide, Die, Process Immune Damage.
- **Agent Goal:** Survive and proliferate under environmental and immune pressure.

Cytotoxic T Cell

- **Internal state:** Activation, Exhaustion, Age, Inbox.
- **Perception:** Nearby tumor cells, Oxygen level, Macrophage signals, Tumor suppression signals, Hypoxia signals.
- **Action/Rules:** Move toward tumor cells, Follow cytokine signals, Request macrophage help, Attack tumor cells, Increase/decrease activation, Accumulate exhaustion, Die.
- **Agent Goal:** Find and eliminate tumor cells while maintaining sufficient activation.

Macrophage

- **Internal state:** Phenotype (M1/M2), Age, Inbox.
- **Perception:** Nearby tumor cells, Oxygen level, T-cell help requests.
- **Action/Rules:** Switch between M1 and M2 phenotype, Move toward tumor cells, Activate T cells, Suppress T cells, Support tumor cells, Respond to assistance requests, Die.
- **Agent Goal:** Adapt to the local tumor microenvironment and regulate the immune response.

Environment Agent

- **Internal state:** Inbox / environmental monitoring state.
- **Perception:** Local oxygen levels and the location of tumor, T cells, and macrophages.
- **Action/Rules:** Detect hypoxic regions, Send hypoxia signals to affected cells, Coordinate environmental notifications.

- **Agent Goal:** Monitor environmental stress and communicate relevant conditions to biological agents.

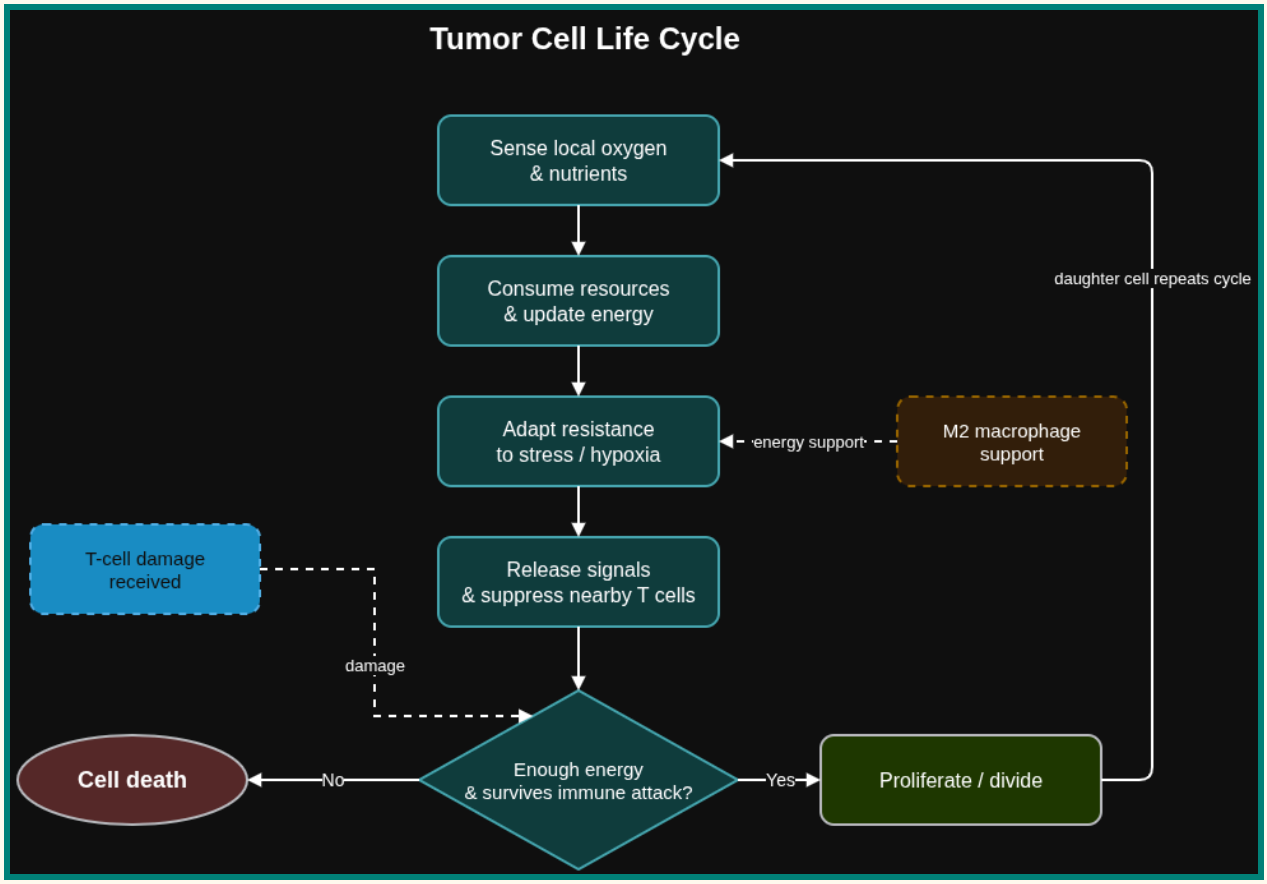


Figure 1: The Tumor Cell Life Cycle

EXPERIMENTATION (WIP)

1. OXYGEN DEPRIVATION

- Setup: Oxygen input is set to 0. Mean-tumor-resistance is 0.0499 and 60 tumor cells
- Results: Cells die and proliferation slow down, but in some cases tumor cells continue to resist and adapt to low Oxygen levels [2], in other cases the tumor is eradicated. If tumor cells survive the initial hypoxia conditions, their resistance increases due to the inverse relationship between hypoxia and resistance, making the cells progressively more difficult to eliminate [1].

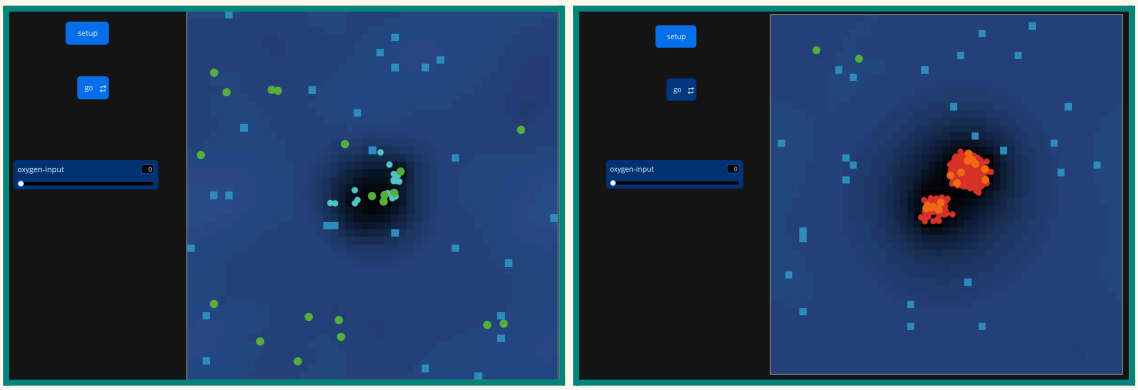


Figure 1: Tumor Eradication Scenario / Figure 2: Tumor Adaptation and Survival Scenario

2. HIGH IMMUNITY

- Experiment: Immune strength is set to high.
- Results: Tumor cells die faster and the number of immune-kills rises compared to baseline [4], for substantial levels of immune strength '>1', each simulation terminates with the tumor eradication scenario [5]

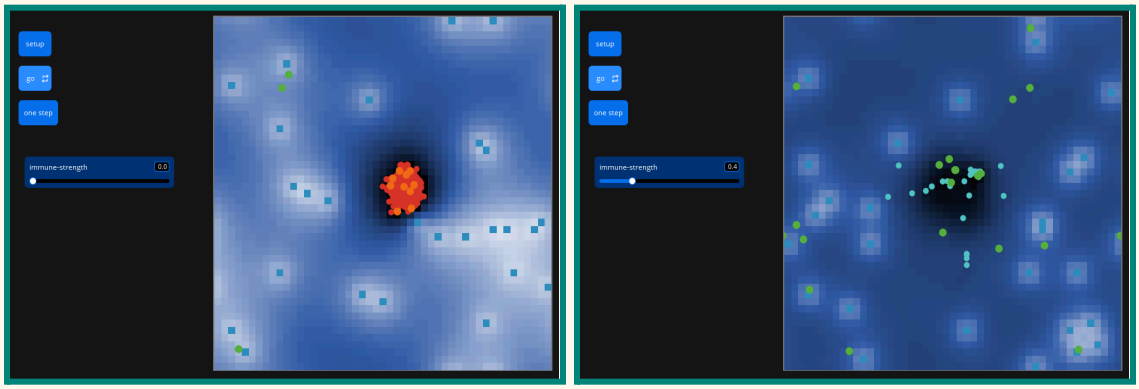


Figure 3: Minimum Immune Strength / Figure 4: *Baseline Immune Strength*

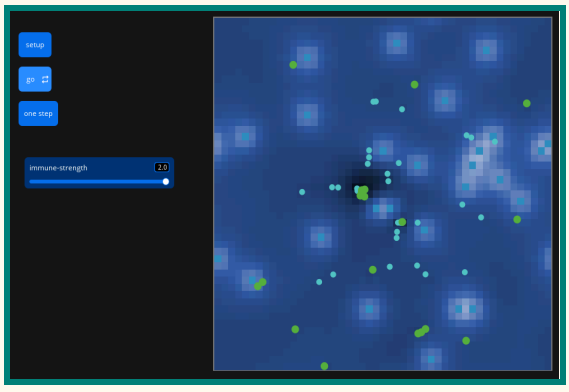


Figure 5: Maximum Immune Strength